CST8506 - Lab 5
Clustering, Outlier Detection & Stacking

Due Date: Check Brightspace for due dates.

Introduction

The goal of this lab is to:

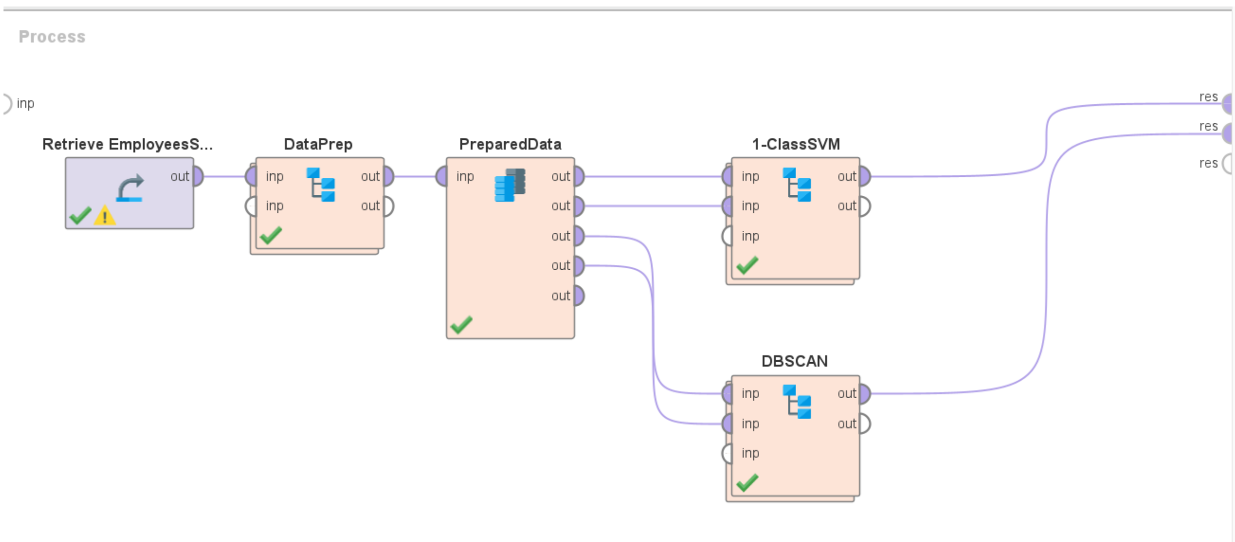
- find outliers in the EmployeesSalary file using 1-class SVM and perform clustering using DBSCAN.
- handle the imbalance in Diabetes dataset (you have the arff file in Weka's data folder) using resampling techniques and compare the classification results before and after resampling.

All tasks should be performed using RapidMiner. Compare your results with the results that you got with kMeans, LOF and ISF earlier.

Steps (all these steps should be done in RapidMiner as one RMP file):

Task 1: Clustering & Outlier Detection

You must have your process for this task as shown below:



1. For the data preparation, you must select relevant columns and do the required data preparation steps. Keep in mind that you will be using distance-based approach, so your data should be prepared accordingly. For distance-based methods, numerical attributes must be scaled, and nominal attributes must be one-hot encoded.
2. You must find outliers using 1-class SVM approach. This operator will give you an outlier score. You must convert the score to a flag. Then join the results with the original dataset and select the original columns along with the result columns. When you select salary, make sure to select the original salary.

3. Take a screenshot of the outlier instances. It should have the format:

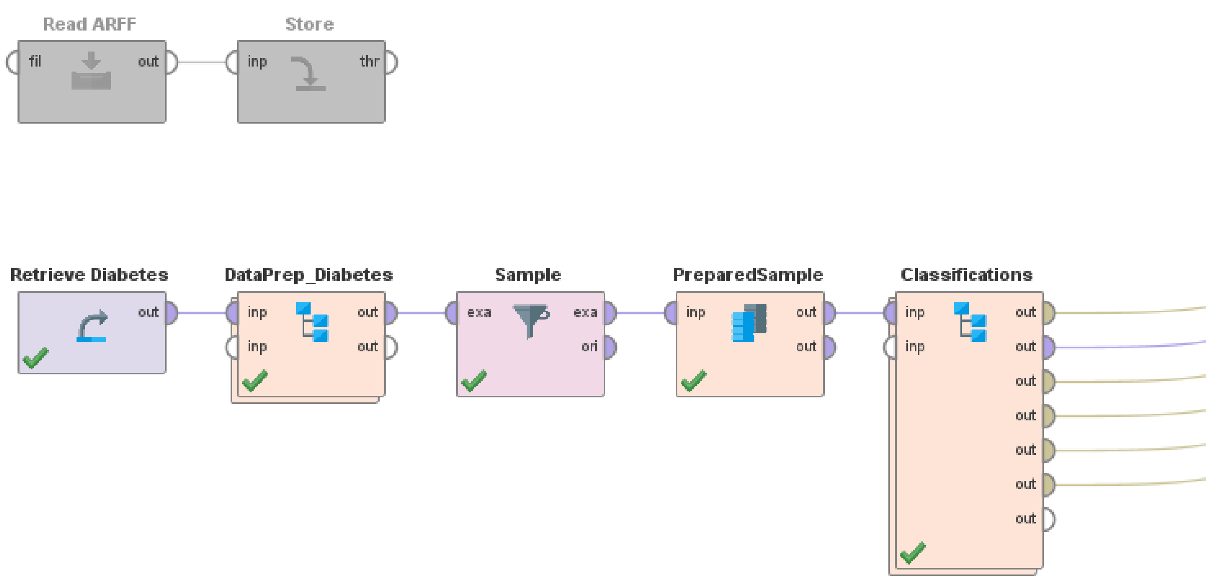
Row No.	Id	outlier	outlier_flag ↓	first_name	last_name	email	Address	Country	Branch	Currency	OrigSalary

4. Now, perform clustering using DBSCAN approach and join the results with the original data and finally select relevant original columns along with the result columns.
5. Take a screenshot of the model. Also, take a screenshot of the noise instances. Format:

Row No.	Id	score(cluster_0)	score(cluster_1)	score(cluster_2)	score(cluster_3)	cluster ↑	first_name	last_name	email	Address	Country	Branch	Currency	OrigSalary

Task 2: Sampling and Stacking

You must have your process for this task as shown below:



1. Read the diabetes file (you have the arff file in Weka's data folder) and store it in RapidMiner's Data under Local Repository.
2. Retrieve the dataset, set the class column and perform any required data preparation steps. Make sure to prepare the dataset for distance-based approaches.
3. Perform resampling on the prepared data to resolve the class imbalance problem.
4. After multiplying the dataset, apply a 70-30 split to split the data into train and test data, again multiply both train and test sets. This is to make sure that you will be using the same data for various modeling. Test data will also be the same for all models.
5. You must use kNN, Naïve Bayes, SVM and logistic regression models. Apply each model on the test set and get the results.
6. Now, create a stacked model with kNN, NB, and SVM as base learners and Logistic regression as the stacking model. Apply the model on the test set and get the results.
7. Tabulate all results in a table – this should be done manually.

In order to get grades:

1. You should be ready with your RapidMiner process and results.
2. Submit your .rmp file AND the answer document to Brightspace. Missing any of these will result in 0 grade.
3. Don’t zip, zipped file will NOT be graded.